

Mathematical Methods I

Differential forms and the exterior derivative

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1. Differential forms and degree

Differential forms can be evaluated on vectors, pairs of vectors, and, more generally, collections of tangent vectors, while keeping track of orientation. In these notes, we work with smooth forms on an open set $\Omega \subseteq \mathbb{R}^n$.

Definition 1.1 (Differential k -form)

For $k \geq 1$, a k -form ω assigns to each point p an alternating multilinear map

$$\omega_p : \underbrace{T_p\Omega \times \cdots \times T_p\Omega}_{k \text{ factors}} \longrightarrow \mathbb{R}.$$

Multilinearity means linearity in each argument. Alternation means that interchanging two vectors changes the sign and that the value is zero whenever two arguments coincide. A **0-form** is a smooth scalar function.

Degrees in $\mathbb{R}^3$

$$\begin{aligned} \text{0-form:} & \quad f(x, y, z), \\ \text{1-form:} & \quad A_x dx + A_y dy + A_z dz, \\ \text{2-form:} & \quad F_x dy \wedge dz + F_y dz \wedge dx + F_z dx \wedge dy, \\ \text{3-form:} & \quad K dx \wedge dy \wedge dz. \end{aligned}$$

The coefficients are functions of the point. The 1-forms serve as line integrands, the 2-forms as oriented surface integrands, and the 3-forms as oriented volume integrands.

General coordinate expression

A k -form can be written uniquely as

$$\omega = \sum_{i_1 < \cdots < i_k} a_{i_1 \dots i_k}(x) dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

The sum uses only strictly increasing indices. In dimension n , this basis has $\binom{n}{k}$ elements, and every form of degree $k > n$ is zero.

In the abbreviated notation $\omega = \sum_I a_I dx^I$, the multi-index $I = (i_1, \dots, i_k)$ is ordered and $dx^I = dx^{i_1} \wedge \cdots \wedge dx^{i_k}$.

2. Wedge product and orientation

The **wedge product** $\wedge$ combines a k -form and an ℓ -form to produce a $(k + \ell)$ -form. It is bilinear, associative, and satisfies

$$\omega \wedge \eta = (-1)^{k\ell} \eta \wedge \omega.$$

For 1-forms α, β , this gives $\alpha \wedge \beta = -\beta \wedge \alpha$ and $\alpha \wedge \alpha = 0$. In particular, $dx \wedge dx = 0$: any product of coordinate differentials containing a repeated factor vanishes.

Evaluation of two 1-forms

For vectors u, v based at the same point,

$$(\alpha \wedge \beta)(u, v) = \alpha(u)\beta(v) - \alpha(v)\beta(u).$$

Thus, $(dx \wedge dy)(u, v) = u^x v^y - u^y v^x$ is the signed area of the projections of u, v onto the xy -plane.

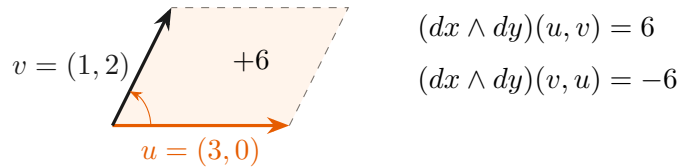


Figure 1: The order (u, v) gives the positive orientation. Interchanging the vectors preserves the geometric area and changes its sign.

Oriented volume

The Euclidean volume form satisfies

$$(dx \wedge dy \wedge dz)(u, v, w) = \det \begin{pmatrix} u^x & v^x & w^x \\ u^y & v^y & w^y \\ u^z & v^z & w^z \end{pmatrix}.$$

Its absolute value is the volume of the parallelepiped spanned by the three vectors; the sign records its orientation.

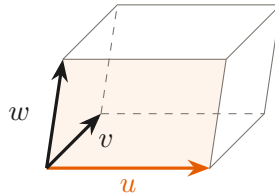


Figure 2: Projection of a parallelepiped. The determinant is calculated using the vectors in space, not their projections in the drawing.

3. Wedge product and Jacobian

Let $x = x(u, v)$, $y = y(u, v)$ be a locally invertible change of coordinates. By the chain rule,

$$dx = x_u du + x_v dv, \quad dy = y_u du + y_v dv,$$

where $x_u = \partial x / \partial u$, and similarly for the other coefficients.

Expansion in two dimensions

We expand without changing the order of the differentials:

$$\begin{aligned} dx \wedge dy &= (x_u du + x_v dv) \wedge (y_u du + y_v dv) \\ &= x_u y_u du \wedge du + x_u y_v du \wedge dv \\ &\quad + x_v y_u dv \wedge du + x_v y_v dv \wedge dv. \end{aligned}$$

The repeated terms vanish and $dv \wedge du = -du \wedge dv$. Therefore,

Theorem 3.1 (Transformation of the oriented area element)

$$dx \wedge dy = (x_u y_v - x_v y_u) du \wedge dv = \det(J) du \wedge dv, \quad J = \begin{pmatrix} x_u & x_v \\ y_u & y_v \end{pmatrix}.$$

The determinant appears because the wedge product is alternating.

Example 3.2 (Polar coordinates)

With $x = r \cos \theta$, $y = r \sin \theta$, $r > 0$, and a local choice of angle,

$$dx = \cos \theta dr - r \sin \theta d\theta, \quad dy = \sin \theta dr + r \cos \theta d\theta.$$

Expanding and eliminating the repeated terms,

$$\begin{aligned} dx \wedge dy &= r \cos^2 \theta dr \wedge d\theta - r \sin^2 \theta d\theta \wedge dr \\ &= r(\cos^2 \theta + \sin^2 \theta) dr \wedge d\theta \\ &= r dr \wedge d\theta. \end{aligned}$$

The sign of the second term changes when $d\theta \wedge dr$ is reordered.

Determinant and absolute value

Forms record orientation and transform by $\det J$. When changing variables in an integral with respect to unoriented area or volume, $|\det J|$ appears instead. For example, interchanging u and v changes the sign of $du \wedge dv$.

4. The Jacobian in dimension n

Let $x^1, \dots, x^n$ be functions of new coordinates $u^1, \dots, u^n$, with invertible Jacobian

$$J^i_j = \frac{\partial x^i}{\partial u^j}, \quad dx^i = \sum_{j=1}^n J^i_j du^j.$$

Theorem 4.1 (Transformation of a top-degree form)

$$dx^1 \wedge \dots \wedge dx^n = \det(J) du^1 \wedge \dots \wedge du^n.$$

Proof using the Levi-Civita symbol

The completely antisymmetric symbol $\varepsilon_{j_1 \dots j_n}$ is defined by

$$\varepsilon_{j_1 \dots j_n} = \begin{cases} +1, & (j_1, \dots, j_n) \text{ is an even permutation of } (1, \dots, n), \\ -1, & (j_1, \dots, j_n) \text{ is an odd permutation,} \\ 0, & \text{there are repeated indices.} \end{cases}$$

The alternating property of the wedge product is then expressed as

$$du^{j_1} \wedge \dots \wedge du^{j_n} = \varepsilon_{j_1 \dots j_n} du^1 \wedge \dots \wedge du^n.$$

Substituting $dx^i = \sum_{j_i=1}^n J^i_{j_i} du^{j_i}$ and expanding,

$$\begin{aligned} dx^1 \wedge \dots \wedge dx^n &= \sum_{j_1, \dots, j_n=1}^n J^1_{j_1} \dots J^n_{j_n} du^{j_1} \wedge \dots \wedge du^{j_n} \\ &= \left(\sum_{j_1, \dots, j_n=1}^n \varepsilon_{j_1 \dots j_n} J^1_{j_1} \dots J^n_{j_n} \right) du^1 \wedge \dots \wedge du^n. \end{aligned}$$

Terms with repeated indices vanish. The remaining terms correspond to permutations $\sigma \in S_n$, so the coefficient is

$$\sum_{\sigma \in S_n} \varepsilon_{\sigma(1) \dots \sigma(n)} \frac{\partial x^1}{\partial u^{\sigma(1)}} \dots \frac{\partial x^n}{\partial u^{\sigma(n)}} = \det \left(\frac{\partial x^i}{\partial u^j} \right).$$

Substituting this determinant into the preceding expansion proves the theorem.

Example 4.2 (Cylindrical coordinates)

For $x = r \cos \theta$, $y = r \sin \theta$, $z = z$, with $r > 0$, the Jacobian has the polar block and a 1 in the z direction. Thus, $\det J = r$ and

$$dx \wedge dy \wedge dz = r dr \wedge d\theta \wedge dz.$$

5. The exterior derivative

Definition 5.1 (The operator d)

The **exterior derivative** raises the degree of a form by one:

$$d : \Omega^k(\Omega) \longrightarrow \Omega^{k+1}(\Omega).$$

Here $\Omega^k(\Omega)$ denotes the space of smooth k -forms. In coordinates,

$$d\left(\sum_I a_I dx^I\right) = \sum_I da_I \wedge dx^I, \quad da_I = \sum_j \frac{\partial a_I}{\partial x^j} dx^j.$$

The coefficients are differentiated, and their differentials are placed to the left of the exterior basis, preserving its order.

For a function, this definition agrees with $df = \sum_j f_{x^j} dx^j$. For the coordinate basis forms, $d(dx^i) = 0$.

Derivative of a 1-form

If $\alpha = \sum_i \alpha_i dx^i$, then

$$d\alpha = \sum_{i,j} \frac{\partial \alpha_i}{\partial x^j} dx^j \wedge dx^i.$$

In $\mathbb{R}^3$, for $\alpha = P dx + Q dy + R dz$,

$$\begin{aligned} d\alpha &= (R_y - Q_z) dy \wedge dz \\ &\quad + (P_z - R_x) dz \wedge dx \\ &\quad + (Q_x - P_y) dx \wedge dy. \end{aligned}$$

Here $R_y = \partial R / \partial y$, and similarly for the other partial derivatives. The signs arise from reordering products of differentials.

Example 5.2 (Sign calculation)

Let $\alpha = xy dx + x^2 dy$. Since $d(xy) = y dx + x dy$ and $d(x^2) = 2x dx$, we obtain

$$\begin{aligned} d\alpha &= (y dx + x dy) \wedge dx + 2x dx \wedge dy \\ &= x dy \wedge dx + 2x dx \wedge dy \\ &= (-x + 2x) dx \wedge dy = x dx \wedge dy. \end{aligned}$$

The term $y dx \wedge dx$ is zero. Reordering $dy \wedge dx$ produces the minus sign.

6. Linearity and product rules

The operator d is linear over constants: $d(a\omega + b\eta) = a d\omega + b d\eta$ for forms of the same degree and $a, b \in \mathbb{R}$. When the factor is a function, there is an additional term.

Proposition 6.1 (Functions and forms)

For functions f, g and a form α ,

$$d(fg) = g df + f dg, \quad \boxed{d(f\alpha) = df \wedge \alpha + f d\alpha.}$$

In the second formula, if α is a 1-form, both terms are 2-forms.

Proof for functions

The product rule for partial derivatives gives

$$d(fg) = \sum_i (g \partial_i f + f \partial_i g) dx^i = g df + f dg.$$

Proof for a function times a 1-form

If $\alpha = \sum_i \alpha_i dx^i$, then

$$\begin{aligned} d(f\alpha) &= \sum_i d(f\alpha_i) \wedge dx^i \\ &= \sum_i (\alpha_i df + f d\alpha_i) \wedge dx^i \\ &= df \wedge \left(\sum_i \alpha_i dx^i \right) + f \left(\sum_i d\alpha_i \wedge dx^i \right) \\ &= df \wedge \alpha + f d\alpha. \end{aligned}$$

Example 6.2 (Both terms are necessary)

Let $f = z$ and $\alpha = x dy$. Then $df = dz$ and $d\alpha = dx \wedge dy$. The product rule gives

$$\begin{aligned} d(f\alpha) &= dz \wedge (x dy) + z dx \wedge dy \\ &= -x dy \wedge dz + z dx \wedge dy. \end{aligned}$$

Here both terms contribute. Directly, $f\alpha = xz dy$, so

$$\begin{aligned} d(xz dy) &= (z dx + x dz) \wedge dy \\ &= z dx \wedge dy - x dy \wedge dz. \end{aligned}$$

Both methods give the same result.

Function, 1-form, and 2-form

In $d(fg)$, the terms $g df$ and $f dg$ are functions times 1-forms. In $d(f\alpha)$, the term $df \wedge \alpha$ uses the wedge product. Keeping the degrees visible helps distinguish these operations.

7. The graded Leibniz rule

Theorem 7.1 (Derivative of a wedge product)

If ω has degree k and η has degree ℓ , then

$$d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta.$$

The sign depends on the degree of the first factor. For $k = 0$, this recovers the rule for a function times a form; for $k = 1$, a minus sign appears.

Proof and origin of the sign

By linearity, it is enough to consider monomials $\omega = A dx^I$ and $\eta = B dx^J$, with $|I| = k$ and $|J| = \ell$. Then

$$\begin{aligned} d(\omega \wedge \eta) &= d(AB) \wedge dx^I \wedge dx^J \\ &= B dA \wedge dx^I \wedge dx^J + A dB \wedge dx^I \wedge dx^J. \end{aligned}$$

The first term is $d\omega \wedge \eta$. To identify the second, we move the 1-form dB past the k factors of dx^I :

$$dB \wedge dx^I = (-1)^k dx^I \wedge dB.$$

Each interchange introduces a minus sign. Therefore,

$$A dB \wedge dx^I \wedge dx^J = (-1)^k (A dx^I) \wedge (dB \wedge dx^J) = (-1)^k \omega \wedge d\eta.$$

Adding the terms gives the identity for monomials and, by expansion, for general forms.

Example 7.2 (Product of two 1-forms)

In $\mathbb{R}^3$, let $\omega = x dy$ and $\eta = z dx$. Their derivatives are

$$d\omega = dx \wedge dy, \quad d\eta = dz \wedge dx.$$

Since $\deg \omega = 1$,

$$\begin{aligned} d(\omega \wedge \eta) &= (dx \wedge dy) \wedge (z dx) - (x dy) \wedge (dz \wedge dx) \\ &= -x dy \wedge dz \wedge dx = -x dx \wedge dy \wedge dz. \end{aligned}$$

The first term vanishes because dx is repeated. In the last step, moving dx to the beginning requires two interchanges, so the sign is preserved.

Direct check: $\omega \wedge \eta = -xz dx \wedge dy$, so

$$d(\omega \wedge \eta) = -(z dx + x dz) \wedge dx \wedge dy = -x dx \wedge dy \wedge dz.$$

8. Nilpotence: $d^2 = 0$

Theorem 8.1 (Two exterior derivatives)

For every smooth form ω ,

$$d(d\omega) = 0.$$

This property is called the **nilpotence** of d . For the proof, it is enough that the coefficients have continuous partial derivatives up to second order.

First: a scalar function

If f is of class C^2 , then

$$df = \sum_i \partial_i f dx^i, \quad d(df) = \sum_{j,i} \partial_j \partial_i f dx^j \wedge dx^i.$$

The terms with $i = j$ are zero. Grouping the remaining terms in pairs,

$$d(df) = \sum_{j < i} (\partial_j \partial_i f - \partial_i \partial_j f) dx^j \wedge dx^i = 0.$$

Equality of the mixed partial derivatives cancels the coefficients, while antisymmetry provides the minus sign.

Next: a form of any degree

For $\omega = A dx^I$, we have $d\omega = dA \wedge dx^I$. The graded rule gives

$$d(d\omega) = d(dA) \wedge dx^I - dA \wedge d(dx^I) = 0.$$

The first term is zero by the case of functions. The second is also zero because every coordinate differential has zero exterior derivative and, by the graded rule, $d(dx^I) = 0$. Linearity extends the result to any sum of monomials.

Closed and exact forms

A form ω is **closed** if $d\omega = 0$. A k -form, with $k \geq 1$, is **exact** if there exists a $(k-1)$ -form α such that $\omega = d\alpha$.

$$\omega \text{ exact} \implies d\omega = d(d\alpha) = 0 \implies \omega \text{ closed}.$$

The converse requires conditions on the domain.

Example 8.2 (An exact form is closed)

Let $\omega = y dx + x dy = d(xy)$. Directly,

$$d\omega = dy \wedge dx + dx \wedge dy = 0.$$

The cancellation illustrates $d^2(xy) = 0$.

9. Poincaré lemma: from closed to exact

Theorem 9.1 (Poincaré lemma)

On a star-shaped open subset of $\mathbb{R}^n$, every smooth closed form of degree $k \geq 1$ is exact. In particular, every closed form of positive degree is **locally exact**, because every point has a ball contained in the domain.

An open set is star-shaped with respect to the origin if it contains the entire line segment tx , $0 \leq t \leq 1$, for every x in the domain. The same exactness conclusion holds on contractible open sets; the construction below treats the star-shaped case.

Radial construction of a primitive

Define $\varphi_t(x) = tx$. For $\omega = \sum_I a_I(x) dx^I$ of degree $k \geq 1$, the homotopy operator K lowers the degree by one:

$$K\omega = \int_0^1 t^{k-1} \sum_{j,I} x^j a_I(tx) \iota_{\partial_j}(dx^I) dt.$$

Here ι_{∂_j} is contraction: it inserts ∂_j as the first argument of the form. For example,

$$\iota_{\partial_x}(dx \wedge dy) = dy, \quad \iota_{\partial_y}(dx \wedge dy) = -dx.$$

Homotopy identity and conclusion

Integrating the variation of $\varphi_t^* \omega$ from $t = 0$ to $t = 1$ gives

$$\omega - \varphi_0^* \omega = d(K\omega) + K(d\omega).$$

The symbol φ_t^* denotes the pullback of the form by the map $x \mapsto tx$. For $k \geq 1$, the constant map φ_0 has zero differential, so $\varphi_0^* \omega = 0$. If, in addition, $d\omega = 0$,

$$\omega = d(K\omega).$$

Thus, $\alpha = K\omega$ is a primitive of ω .

This identity is obtained by applying the chain rule and Cartan's formula for the variation of a form along the homotopy. The factor t^{k-1} accounts for the scaling of its $k-1$ remaining arguments after contraction.

Example 9.2 (A primitive of $dx \wedge dy$)

For $\omega = dx \wedge dy$, the preceding formula gives

$$K\omega = \int_0^1 t(x dy - y dx) dt = \frac{1}{2}(x dy - y dx).$$

Indeed,

$$d(K\omega) = \frac{1}{2}(dx \wedge dy - dy \wedge dx) = dx \wedge dy.$$

10. Local and global scope of the lemma

An explicit primitive for 1-forms

If $\omega = \sum_i a_i(x) dx^i$ is closed on an open set that is star-shaped with respect to the origin, the case $k = 1$ of the radial construction produces the function

$$f(x) = K\omega = \int_0^1 \sum_i x^i a_i(tx) dt.$$

We can verify $df = \omega$ without using contraction notation. The closedness condition implies $\partial_j a_i = \partial_i a_j$, so

$$\begin{aligned} \partial_j f(x) &= \int_0^1 \left[a_j(tx) + t \sum_i x^i \partial_j a_i(tx) \right] dt \\ &= \int_0^1 \left[a_j(tx) + t \sum_i x^i \partial_i a_j(tx) \right] dt \\ &= \int_0^1 \frac{d}{dt} (t a_j(tx)) dt = a_j(x). \end{aligned}$$

Therefore, $df = \sum_j a_j dx^j = \omega$.

Example 10.1 (Closed, but not globally exact)

On $\Omega = \mathbb{R}^2 \setminus \{0\}$, consider

$$\omega = \frac{-y dx + x dy}{x^2 + y^2}.$$

If $P = -y/(x^2 + y^2)$ and $Q = x/(x^2 + y^2)$, then

$$Q_x = P_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad d\omega = (Q_x - P_y) dx \wedge dy = 0.$$

However, for the circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, we obtain

$$\omega_{\gamma(t)}(\gamma'(t)) = 1, \quad \int_{\gamma} \omega = \int_0^{2\pi} 1 dt = 2\pi.$$

If $\omega = df$ globally, the chain rule would give

$$\int_{\gamma} df = f(\gamma(2\pi)) - f(\gamma(0)) = 0,$$

a contradiction. The form is locally $d\theta$, but there is no single smooth real-valued angle on the entire punctured plane that serves as its potential.

Two different statements

The nilpotence $d^2 = 0$ guarantees that every exact form is closed. The Poincaré lemma guarantees the local converse in positive degrees. In degree zero, $df = 0$ means that f is locally constant; the homotopy identity takes the form $f - f(0) = K(df)$ on the star-shaped domain.